

RENDELL ROCKY PADU | Full Stack Engineer

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SUMMARY

A Modern-era Full Stack Developer with a strong foundation in Data engineering, ML and leveraging Generative AI to modernize workflows. Experienced in building web platforms with dynamic frontends, scalable backends and designing real-time data pipelines to extract actionable insights. Committed to delivering innovative, data-informed solutions in fast-paced, collaborative environments.

EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering **August 2023 - May 2025**
Master of Science, Computer Science, Artificial Intelligence Track (GPA: 3.63/4.00)
• **Coursework:** Social Media Data Pipeline, Operating Systems, Machine Learning, Database Management System

University of Mumbai, Mumbai, India **August 2019 - May 2023**
Bachelor of Technology, Electronics and Computer Science
Mumbai, MH, India

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C, Bash
- **Data Engineering & Big Data:** PySpark, Databricks, BigQuery, TimescaleDB, PostgreSQL, MongoDB, NoSQL
- **Data Science & Analytics:** Advanced Excel, Power BI, Tableau, Google Cloud Compute, Chart.js, Data analysis, Data analytics, Data transformation, Data modeling, Data architecture, Statistical methodologies, Linear algebra, Bayesian statistics, Group theory, AI techniques, Data science tools, TensorFlow
- **Web Development & APIs:** HTML, CSS, JavaScript, React.js, Flask, REST APIs, OAuth2, Faktery, VADER Sentiment Analysis
- **Cloud & DevOps:** Google Cloud Platform (GCP), AWS, Docker, Git, GitHub, Linux

PROFESSIONAL EXPERIENCE

Data Analyst | Violence Abuse and Rape Crisis Centre (VARCC) | *Binghamton, NY* **August 2024 - Present**

- Assisted visitors while managing data collection and analysis using Excel to evaluate datasets, generate operational insights, and support gender violence prevention initiatives
- Produced structured reports and interactive visualizations with Power BI and Excel, highlighting trends in safety incidents and service utilization to enable data-driven decision-making
- Applied the data pipeline project to extract insights from Reddit on domestic violence discussions, applying data processing and analytical techniques to refine crisis management strategies

Full Stack Developer | Binghamton University Office of Sustainability | *Binghamton, NY* **May 2024 - August 2024**

- Devised ReactJS and Flask based web application to streamline environmental initiatives and resource management
- Instituted Excel imports for bulk data upload, cutting inspection time from 5 to 3 hours
- Co-developed an admin panel with role-based access controls and utilized Firebase and Google Cloud for deployment

Freelancer, Web Developer | Gloria Mahila Mandal | *Mumbai, India* **February 2022 - May 2022**

- Crafted a static website using HTML, CSS, and Bootstrap that effectively communicated the organization's key initiatives for women's empowerment and community service
- Implemented a user-friendly web interface to showcase organizational initiatives, resulting in increased participant engagement at events

PROJECT EXPERIENCE

Real Time Data Pipeline for Binghamton University Dean of Students Office | *Data Engineer* **August 2024 - December 2024**

- Constructed a real-time data pipeline to continuously collect and monitor discussions on domestic violence and sexual assault from data collection techniques, including OAuth2 authentication for Reddit and direct HTTP requests for 4chan's public APIs
- Orchestrated a scalable producer-consumer architecture using Faktery to manage and distribute data collection tasks across multiple workers and used TimescaleDB for storage and retrieval of the massive dataset, which totaled over 4.6 million posts
- Integrated sentiment analysis using the VADER algorithm and toxicity measurement via the ModerateHatespeech API to examine emotional patterns and harmful content across the platforms
- Analyzed discussion volume, sentiment, toxicity, and temporal trends across platforms, delivering insights to the Dean of Students Office to support research and enhance crisis management initiatives

Uber like Data Analysis | *Independent Project* **December 2024**

- Engineered a data model to analyze Uber ride data, uncovering trends and insights related to user behavior and ride patterns
- Executed custom Python scripts using Pandas and NumPy to clean missing values, normalize timestamps, and calculate ride durations and peak hours; deployed the scripts on a Google Cloud Compute instance to process over 10 million records efficiently
- Built an interactive dashboard using Mage.ai and BigQuery, enabling users to explore key metrics and visualize trends

Amazon Kindle Review Analysis | *Database management Engineer* **November 2024**

- Designed Flask REST API with efficient query endpoints, implementing pagination and response caching to handle high-volume review analysis requests and structured JSON responses to minimize frontend data processing
- Developed an interactive dashboard leveraging Chart.js to create visualizations, including bar charts and line graphs for review counts by ratings, sentiment distribution and trend analysis, simplifying complex data
- Optimized NoSQL queries in MongoDB by indexing frequently searched fields (e.g., review ratings, keywords) and utilizing aggregation pipelines, reducing query execution time and enhancing data retrieval efficiency